MIL-P-23004A (Wep)
13 March 1964

SUPERSEDING
MIL-P-23004 (Wep)
21 August 1961

MILITARY SPECIFICATION

POD (FOR GUN POD MARK 4 MOD 0)

1. SCOPE

1.1 Scope. This specification covers Pod (for Gun Pod Mark 4 Mod 0), referred to herein as the pod.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of.
MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-P-5518	Pneumatic Systems, Aircraft; Design, Installation, and Data Requirements for.
MIL-T-5522	Test Procedures for Aircraft Hydraulic and Pneumatic Systems, General.
MIL-E-7080	Electric Equipment, Piloted Aircraft, Installation and Selection of, General Specification for.
MIL-A-8591	Airborne Stores and Associated Equipment; General Design Criteria for.

FSC 1005

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-143	Specifications and Standards, Order of Precedence for Selection of.
MIL-STD-704	Electric Power, Aircraft; Characteristics and Utilization of.

DRAWINGS

Bureau of Naval Weapons
(Code Ident 10001)

LD 416841	Container, Shipping and Storage, for Tail Assembly.
LD 416842	Container, Shipping and Storage, for Nose Assembly.
LD 416846	Container, Shipping and Storage, for Reservoir Assembly, Feed Chutes, and Ejection Tubes.
LD 416850	Container, Shipping and Storage, for Gun Pod Body.
LD 522078	Gun Mechanism, 20MM, Mark 11 Mod 5.
LD 522079	Loader, Gun, 20MM, Mark 2 Mod 1.
LD 539673	Pod (Assembly).
LD 615425	Link, Ammunition, Mark 6 Mod 4.
LD 615432	Kit, Gun Pod Accessories.
LD 615434	Link, Ammunition, Leading, Mark 6 Mod 5.
LD 615435	Link, Ammunition, Trailing, Mark 6 Mod 6.
982519	Target Practice Round, 20MM, Mark 105 Mod 0.
1445196	Dummy Round, 20MM, Mark 104 Mod 0.

PUBLICATIONS

Military

Military Handbook H1	Cross-Index of Chemically Equivalent Specifications and Identification Code.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

Bureau of Naval Weapons
(Code Ident 10001)

OP 2719	Gun Pod Mark 4 Mod 0; Description, Operation, and Maintenance.
QAP 2071148	Quality Assurance Provisions for Gun Pod Mark 4 Mod 0.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of pods shall be delivered for testing at a facility designated in the contract or order (see 6.2). The preproduction sample shall be manufactured using the methods proposed for production. Any production by the supplier prior to approval of the preproduction sample shall be at the supplier's risk.

3.2 Compliance with documents. Unless otherwise specified, the pod shall be in accordance with the requirements specified herein and in the applicable documents listed in section 2.

3.3 Materials. Materials used in the manufacture of the pod shall conform to the material specifications in the applicable documents listed in section 2. Subject to the approval of the Government representative, alternate materials may be selected in accordance with Military Handbook H1 in the order of precedence specified in Standard MIL-STD-143.

3.4 Design. The design of the pod shall be in accordance with the requirements specified herein and in the applicable documents listed in section 2. The pod shall be assembled with other components of the Mark 4 Gun Pod and prepared for operation as specified in Publication OP 2719. The pod shall meet the requirements of 3.4.1 through 3.4.5.

3.4.1 Safety. When the switch controlling the gun pod clearing function is actuated, the pod shall supply

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

electric power and compressed air to the gun mechanism and loader to disengage the loader sprocket from the gun mechanism. The pod shall cycle the gun mechanism not fewer than three times in 500 milliseconds, ejecting cartridges or cartridge cases with a velocity sufficient to pass through the loader and the pod ejection tubes without catching or binding.

3.4.2 Function.

3.4.2.1 Manual charging. When the switch controlling the gun pod charging function is actuated, the pod shall cycle the gun mechanism to carry rounds from the ram position to firing position. Any cartridges or cartridge cases at the eject position shall be ejected with a velocity sufficient to clear incoming links and to pass through the ejection tubes without catching or binding.

3.4.2.2 Automatic charging. When the pod electrical system is wired for automatic charging and the gun fails to fire when the firing circuits are energized, the pod shall supply signals and power to the gun for automatic charging within 200 milliseconds. Live rounds shall be cycled into firing position and any rounds or cases at the eject station shall be ejected.

3.4.2.3 Rate of fire. A switch shall be provided in the pod junction box to preselect the rate of fire of the gun pod. The normal rate of fire shall be 4000 rounds per minute plus or minus 200 rounds per minute. The slow rate of fire shall be 700 rounds per minute plus or minus 100 rounds per minute.

3.4.3 Life.

3.4.3.1 Operational life. The pod shall be capable of feeding 130,000 service rounds and firing the same number through the gun mechanism with normal parts replacement and servicing. The parts replacement schedule shall be as specified in Publication OP 2719.

3.4.3.2 Shock and vibration. When preserved and packaged (see section 5), the pod shall withstand shocks, impacts from accidental dropping, and vibrations of logistic handling and transportation without damage.

3.4.3.3 Storage life. When preserved and packaged, the pod shall be capable of storage without deterioration for

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

not less than three years.

3.4.4 Interchangeability. All pod component parts which are not formed into inseparable assemblies shall be capable of assembly without selection or fitting at industrial facilities and at the fleet level.

3.4.5 Coordination.

3.4.5.1 Service ammunition. The pod shall feed, cycle, and fire service ammunition in the Mark 100 series.

3.4.5.2 Aircraft gun. The pod shall assemble with the 20MM Aircraft Gun Mark 11 Mod 5 without selection or fitting.

3.4.5.3 Components. The electrical and pneumatic components of the pod shall be capable of nonselective connection with those of the gun mechanism.

3.5 Construction.

3.5.1 Ejector reinforcement. The ejector reinforcement of the pod shall meet the applicable requirements of figure 4 of Specification MIL-A-8591.

3.5.2 Sway brace reinforcement. The sway brace reinforcement of the pod shall meet the applicable requirements of figure 4 of Specification MIL-A-8591.

3.5.3 Cradling reinforcement. The cradling reinforcement of the pod shall meet the applicable requirements of figure 5 of Specification MIL-A-8591.

3.6 Details of components.

3.6.1 Electrical system.

3.6.1.1 External power. The pod shall conform to Standard MIL-STD-704, and shall meet all other requirements of this specification when furnished with external power as defined in Standard MIL-STD-704 and as follows:

(a) A 28 volt plus or minus 4.0 volt direct current, negative grounded power source capable of delivering not less than 7.0 amperes during charging.

(b) A 120/208 volt, three-phase, 400 cycle alternating current power source capable of delivering not less than 2.0 amperes per phase on two phases during firing or 120 volt single phase at 4.0 amperes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

3.6.1.2 Wiring and installation. The pod shall be capable of meeting the requirements of this specification when wired in accordance with Specification MIL-W-5088 and installed in accordance with Specification MIL-E-7080. The requirements of Specification MIL-B-5087 are not applicable to this specification.

3.6.1.3 Induced voltage. There shall be no induced voltage in the pod electrical system at any time which could energize the firing circuits and accidentally fire ammunition.

3.6.2 Pneumatic system.

3.6.2.1 Tube assemblies. All tube assemblies shall withstand a proof pressure of 6000 psig plus or minus 100 psig and shall meet the requirements of Specification MIL-P-5518.

3.6.2.2 Leakage. When the cutoff valve is closed, pressure loss measured at the reservoir assembly shall not exceed 25 psig in 24 hours. When the cutoff valve is open, pressure loss measured at the reservoir assembly shall not be more than 100 psig in 24 hours.

3.6.2.3 System proof pressure. Upstream of the pressure regulator, the pod system shall withstand a proof pressure (relief reseat pressure) of 3300 psig plus or minus 100 psig. When the pressure regulator is set as specified in Drawing 2094307, the system shall withstand a proof pressure (lockup pressure) downstream of the regulator of 1500 psig maximum.

3.7 Performance characteristics. When tested as specified in 4.6.3.1, the pod shall withstand the firing of the complete contents of four ammunition magazines without a stoppage, and shall meet the requirements of 3.4.2.

3.8 Temperature extremes. The pod shall meet the requirements of 3.4.1 and 3.4.2 at any temperature between minus 65 degrees Fahrenheit (F.) and 165 degrees F.

3.9 Government-furnished property. The following, in quantities specified in the contract or order, are required to determine compliance of the pod with the requirements of this specification:

(a) Dummy Rounds, 20MM, Mark 104 Mod 0.
(b) Target Practice Rounds, 20MM, Mark 105 Mod 0.
(c) Ammunition Links, Mark 6 Mods 4, 5, and 6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

3.10 Government-loaned property. The following are required to test the pod for conformance to the requirements of this specification:

(a) Gun Mechanism, 20MM, Mark 11 Mod 5.
(b) Loader, Gun, 20MM, Mark 2 Mod 1.
(c) Ejector Rack Assembly, Aero 7A-1, or equal.
(d) Installation Kit, Mark 4 Mod 0 Gun Pod (Drawing 2471452), or kit matching (c).
(e) Kit, Gun Pod Accessories.

3.11 Workmanship. The quality of workmanship on pod components and on the assembled pod shall assure conformance to all of the requirements of this specification. The assembly of pod components into the pod shall be as specified in Publication OP 2719.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the pod shall be classified as follows:

(a) Preproduction tests.
(b) Acceptance tests.

4.3 Lot. Lot definition, size, and formation shall be as specified in Standard MIL-STD-105.

4.4 Sampling. Unless otherwise specified herein, the provisions set forth in Standard MIL-STD-105 shall govern the establishment of sampling plans and procedures for inspection by attributes.

4.5 Examination.

4.5.1 Components. Prior to assembly of the pod, it shall be ascertained that all components procured under separate documents have been inspected, tested and accepted in

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

accordance with their respective documents.

4.5.2 Packing and marking. It shall be ascertained that packing and marking of the pod conform to section 5 of this specification.

4.6 Tests.

4.6.1 Acceptance tests. Sample pods, selected in accordance with 4.4, shall be subjected to the acceptance tests specified herein, except that the tests of 4.6.1.3, 4.6.1.4, 4.6.1.5, and 4.6.1.6 shall be performed on each pod component or pod intended for delivery to the Government. Lot quality shall be as specified in 4.6.2.

4.6.1.1 Visual and mechanical inspection. The pod and pod components shall be examined for conformance to 3.2, 3.3, and 3.11.

4.6.1.2 Pneumatic test procedure. The supplier shall prepare a pneumatic test procedure and shall submit test reports in accordance with the requirements of Specification MIL-T-5522.

4.6.1.3 Tubing pressure tests. Prior to assembly into the pod, all tube assemblies shall be subjected to proof pressure tests in a burst chamber approved by the Government representative. The tube assemblies shall meet the requirements of 3.6.2.1. Following the tests, the tube assemblies shall be degreased and dust caps conforming to Specification MIL-C-5501 shall be installed.

4.6.1.4 Leakage tests. With the manually operated valve in the off position, charge the pneumatic reservoir as specified in the instructions shown on the plate attached to the reservoir. After charging, the assembled pod shall be allowed to stand for not less than four hours. Record temperature and pressure at the reservoir at the end of the four hour period. Allow the pod to stand for 24 hours. Read the temperature and pressure at the reservoir. After making a temperature-pressure correction, record the temperature and the corrected pressure. If permitted leakage does not conform to the applicable requirements of 3.6.2.2, apply soap solution to the fittings to determine the leakage source. Recharge the reservoir to the specified pressure and allow the pod to stand for not less than two hours. Record temperature and pressure at the reservoir and open the manually operated valve. Allow

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

the pod to stand for 24 hours and repeat the test procedure specified herein. Subject to the approval of the Government representative, an alternate method of determining leakage may be substituted.

4.6.1.5 System pressure test. The proof pressure requirement of 3.6.2.3 shall be verified at the same time that the tests of 4.6.1.4 are performed. Defective components shall be isolated by application of soap solution and shall be replaced.

4.6.1.6 Clearing test. The gun mechanism and loader shall be installed in the pod as specified in Publication OP 2719. The clearing switch shall be actuated. The pod shall conform to 3.4.1.

4.6.1.7 Function tests. Gun pod components shall be assembled and prepared for operation as specified in Publication OP 2719, except that the magazine shall be loaded with two belts of 130 links in each belt. Each belt shall be composed of one empty Mod 5 leading link, followed by 128 Mod 4 links and one Mod 6 trailing link. The belt intended to be fired through the last fire barrel shall contain 129 target practice rounds. The other belt, beginning approximately at the thirtieth link, shall contain five successive dummy rounds. All other rounds in the belt shall be target practice rounds. Cycle the gun and fire the gun pod in three bursts of approximately 86 rounds per burst, at the same time verifying that the gun is charging automatically each time a dummy round is encountered. The pod shall meet the requirements of 3.4.2 and 3.4.5.1.

4.6.2 Acceptance criteria. Lot quality for the test of 4.6.1.1 shall be judged in accordance with Standard MIL-STD-105, Acceptable Quality Level (AQL) 1.0 percent defective. Failure of the pod to meet the requirements of 3.4.1, 3.4.2, 3.4.5.1, and 3.6.1 shall be cause for rejection. The pod will not be judged defective if it is verified to the satisfaction of the Government representative that a gun pod stoppage or other malfunction was caused by a component other than the pod under test.

4.6.3 Preproduction tests. The preproduction tests shall include all of the acceptance tests of 4.6.1.1 through 4.6.1.7 and the tests specified herein.

4.6.3.1 Endurance firing test. Prior to test, the automatic charge shall be deactivated by removing the jumper

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

from terminal board TB301 located in the pod junction box. The gun pod components shall be assembled and prepared for operation as specified in Publication OP 2719. The ammunition magazine shall be completely filled with Mark 105 target practice rounds. The gun shall be cycled and the rounds shall be fired in bursts of approximately 125 rounds per burst. After the first, second, fourth, and fifth bursts, the assembled gun pod shall be allowed to cool for 3.0 minutes plus or minus 15 seconds. After the third burst and last burst, the gun pod shall be allowed to cool until the entire mass of the gun pod has been stabilized at the ambient temperature of the test facility. Any required maintenance shall be performed as specified in Publication OP 2719. The ammunition magazine shall again be filled and the test shall be repeated three additional times with cooling times and maintenance as specified herein.

4.6.3.2 Temperature extremes test. Mark 6 ammunition links shall be belted into four belts of 25 links, each composed of one empty Mod 5 leading link, followed by 23 Mod 4 links and one Mod 6 trailing link. Mod 4 links and the trailing link shall be loaded with Mark 105 target practice rounds. Two belts shall be loaded into the ammunition magazine of a gun pod assembled and prepared for operation as specified in Publication OP 2719. The assembled gun pod shall be temperature-conditioned at minus 65 degrees F. plus or minus 5.0 degrees F. until all components have been stabilized at the specified temperature. Within 15 minutes after removal of the assembled gun pod from the temperature-conditioning chamber, the rounds shall be fired in a single burst. After firing, any required gun pod maintenance shall be performed as specified in Publication OP 2719. The two remaining belts shall be loaded into the magazine, and the gun pod shall be temperature-conditioned at 165 degrees F. plus or minus 5.0 degrees F. until all components have been stabilized at the specified temperature. Within 15 minutes after the gun pod is removed from the temperature-conditioning chamber, the rounds shall be fired in a single burst.

4.6.4 Preproduction acceptance criteria. Failure of the pod to meet any of the requirements of this specification shall be cause for rejection of the preproduction sample. The pod will not be judged defective if it is verified to the satisfaction of the Government representative that a stoppage or other malfunction was caused by a component other than the pod under test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation application. The preservative for the pod shall be in accordance with the applicable drawings listed in Lists of Drawings (LD) 416841, 416842, 416846, and 416850.

5.1.1.2 Unit packaging. Not applicable.

5.2 Packing. Packing shall be level A, B, or C, as specified.

5.2.1 Level A.

5.2.1.1 Exterior containers. Containers for the pod shall be in accordance with the following:

(a) Container, Shipping and Storage, for Tail Assembly (LD 416841).
(b) Container, Shipping and Storage, for Nose Assembly (LD 416842).
(c) Container, Shipping and Storage, for Reservoir Assembly, Feed Chutes, and Ejection Tubes (LD 416846).
(d) Container, Shipping and Storage, for Gun Pod Body (LD 416850).

5.2.2 Level C.

5.2.2.1 Exterior containers. Pod components (see 5.2.1.1) shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. None, unless otherwise specified.

5.3.2 Normal marking. In addition to the marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The pod covered by this specification is intended to house the 20MM Gun Mechanism Mark 11 Mod 5 and the 20MM Gun Loader Mark 2 Mod 1. The complete assembly is referred to as the Gun Pod Mark 4 Mod 0.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of preproduction sample if one is specified (see 3.1).
(c) Facility designated to evaluate the preproduction sample.
(d) Government-furnished property (see 3.9).
(e) Government-loaned property (see 3.10).
(f) Lot size (see 4.3).
(g) Procedure for resubmittal of rejected lots.
(h) Preservation, packaging, and packing levels.
(i) Special marking of shipping containers if required.
(j) Invocation of Specification MIL-Q-9858.

6.3 System reliability. Extreme care must be exercised in establishing the production system and quality control practices for parts covered by this specification to assure an adequate overall gun system reliability. Each subsystem must have considerably higher individual reliability than the overall 7,000 rounds per stoppage needed for the Mark 4 Gun Pods of which this item is a part. These reliabilities cannot be assured by an after the fact inspection to an AQL and are therefore beyond the written requirement of the specification. However, it has been shown that high reliabilities can be achieved even though inspection AQL's are relatively low (ammunition, for example) if a good production system and control is followed. To illustrate the component requirement, the subsystems must perform to stoppage rates no greater than those listed below if the 7,000 rounds per stoppage reliability level of the Mark 4 Gun Pod is to be maintained.

Item	Stoppage/rounds fired
Ammunition	1/100,000
Link	1/100,000
Loader	1/20,000
Gun	1/20,000
Pod	1/50,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:08Z
Check the source to verify that this is the current version before use.